

Knowledge Mobilization Gaps in the Communication of Ovarian Cancer Detection Without Population Screening

A Scoping Review

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Abstract

Background

There is no validated population-based screening test for ovarian cancer in asymptomatic, average-risk individuals, yet misconceptions among the public and some healthcare providers persist regarding detection, including confusion about the role of the Pap test and the nature of symptom-based evaluation.

Methods

This scoping review followed the Arksey and O'Malley framework and PRISMA-ScR guidance. Searches were conducted between October 2025 and January 2026 in PubMed, CINAHL, and Web of Science, alongside structured grey literature searches. Eligible sources addressed communication related to ovarian cancer detection, including screening limitations, Pap test misconceptions, symptom awareness, help-seeking, and knowledge mobilization. Findings were charted and synthesized descriptively.

Results

The search identified 1,660 records, of which 30 sources were included. Routine ovarian cancer screening was consistently not recommended, with symptom-triggered evaluation identified as the appropriate pathway. However, studies reported persistent misunderstanding among women and the general public of Pap testing and ovarian cancer screening, low symptom awareness, false reassurance, and delays in help-seeking. Identified knowledge mobilization

tools included public-facing materials, educational resources, and clinician guidance, though few sources clearly integrated screening limitations, symptom-based evaluation, and distinctions between ovarian and cervical cancer detection.

Conclusions

The primary challenge is not lack of evidence but inconsistent communication across public and clinical contexts. More coherent and action-oriented knowledge mobilization may improve understanding of screening limitations and support timely symptom-based evaluation.

1 Background and Rationale

Ovarian cancer remains one of the most lethal gynecologic malignancies, largely because it is frequently diagnosed at an advanced stage. Unlike cervical cancer, which benefits from organized population-based screening programs, there is currently no validated screening test for ovarian cancer in average-risk, asymptomatic individuals. Large randomized trials and guideline bodies consistently conclude that population screening using CA-125 (a blood-based tumor marker) and transvaginal ultrasound does not reduce mortality and may result in false-positive findings and unnecessary surgical interventions ((National Institute for Health and Care Excellence [NICE], 2015; American College of Obstetricians and Gynecologists [ACOG], 2017). As a result, ovarian cancer detection relies on symptom-triggered diagnostic evaluation rather than routine screening.

Despite this clear clinical consensus, misconceptions about ovarian cancer screening persist among the general public. A common misunderstanding among women is the belief that the Pap test detects ovarian cancer. Qualitative research conducted across multiple U.S. sites has documented confusion regarding the purpose of the Pap test, with many women assuming it screens for multiple gynecologic cancers (Cooper et al., 2011). Survey-based research similarly demonstrates persistent misunderstanding at the population level. In a UK study using the Ovarian Cancer Awareness Measure, 57.9% of respondents incorrectly believed that a National Health Service ovarian cancer screening programme exists. Furthermore, only few participants were able to identify ovarian cancer symptoms without prompting. This suggests ongoing misunderstanding of ovarian cancer detection and potential confusion with other gynecologic screening (Radu et al., 2023).

These misconceptions may contribute to false reassurance, particularly when individuals equate routine gynecologic care with protection against ovarian cancer. In the absence of effective screening, early detection depends on timely recognition of persistent or concerning symptoms and appropriate diagnostic evaluation within primary care pathways (NICE, 2015; ACOG, 2017). However, the ways in which the absence of screening and the role of symptom-based detection are communicated vary across clinical guidelines, advocacy materials, and public-facing educational resources.

Understanding how ovarian cancer detection limitations are framed and interpreted is therefore a knowledge mobilization challenge that causes a clinical one. Mapping how guidance and communication tools address screening limitations, Pap test misconceptions, and symptom-triggered diagnostic pathways is essential to identifying gaps and informing the development of clearer, patient-centered knowledge mobilization resources.

2 Purpose of the Scoping Review

2.1 Purpose

This review examines how ovarian cancer detection is communicated in the absence of effective routine screening, with attention to symptom-based diagnostic pathways and communication gaps that shape public understanding. Particular focus is given to how screening limitations and Pap test misconceptions are communicated to patients, and to how symptom-triggered investigation is presented in primary care guidance. The review aims to identify recurring themes, areas of consistency, communication gaps, and opportunities for improved knowledge mobilization that may contribute to misunderstanding or delayed diagnosis. Findings from this review are intended to inform the development of a knowledge mobilization toolkit to clarify screening limitations, strengthen symptom communication, and support more effective patient–provider conversations.

2.2 Research Questions:

This scoping review was guided by the following primary question:

What evidence and knowledge mobilization resources address communication of ovarian cancer detection in the absence of population-based screening?

Four sub-questions guided this aim:

- ❑ Communication framing: How is the absence of routine ovarian cancer screening described across public-facing and clinician-facing sources?
- ❑ Pap test misconceptions: How are misconceptions regarding Pap testing and ovarian cancer detection described or addressed?
- ❑ Knowledge mobilization tools: What types of communication or educational resources are used to convey screening limitations and symptom-based pathways?
- ❑ Communication gaps: What gaps or inconsistencies in messaging are identified?

3 Methods

3.1 Review Approach

This scoping review was conducted in accordance with the Arksey and O'Malley framework and reported using PRISMA-ScR guidance. Searches were conducted between October 2025 and January 2026 in PubMed, CINAHL, and Web of Science, alongside structured grey literature searches of selected guideline bodies and cancer organizations. Searches were limited to English-language sources published without a date restriction. Sources were eligible if they addressed communication related to ovarian cancer detection, including screening limitations, Pap test misconceptions, symptom awareness, help-seeking, diagnostic pathways, or knowledge mobilization tools. Findings were charted and synthesized descriptively.

A scoping review approach was selected because the aim of this review was to map the range and nature of evidence and guidance related to ovarian cancer screening communication, rather than to evaluate intervention effectiveness or formally assess study quality. This approach was appropriate for identifying how the topic has been addressed across diverse source types, clarifying key concepts, and identifying gaps in the literature. A protocol was not pre-registered prior to the initiation of this review. While scoping reviews can be registered, this review was undertaken as part of an academic program and registration was not complete in advance.

3.2 Data Sources and Search Strategy

The search strategy focused on communication-related evidence relevant to the review questions. Searches were conducted between October 2025 and January 2026 in MEDLINE (via PubMed), CINAHL (via EBSCOhost), and Web of Science. Searches were limited to English-language records.

A structured grey literature search was also undertaken to identify relevant clinician-facing and public-facing materials not indexed in databases. Grey literature sources were selected purposively based on their relevance to ovarian cancer guidance, patient education, and cancer communication in the review context. These sources included the National Institute for Health and Care Excellence (NICE), the American College of Obstetricians and Gynecologists (ACOG), Ovarian Cancer Canada, and the University Health Network Gynecologic Oncology Program. Relevant sections of each website were manually reviewed, including guideline pages, patient education pages, FAQs, and downloadable resources related to ovarian cancer detection, symptoms, and screening. Citation searching of included studies was also conducted to identify additional relevant sources.

The database search combined terms related to ovarian cancer, screening and early detection, Pap test misconceptions, symptom awareness, diagnostic pathways, help-seeking, health communication, and knowledge mobilization. Searches across databases were structured around the following core concepts and terms, with syntax adapted as needed for each database:

("ovarian cancer"[Title/Abstract])
AND (awareness OR knowledge OR communication OR education OR "health literacy" OR
"help-seeking" OR "public awareness" OR "patient education")
AND (symptom* OR "early detection" OR "diagnostic delay" OR "time to diagnosis" OR
"diagnostic interval" OR "diagnostic pathway")
NOT (biomarker* OR methylation OR genom* OR proteom* OR "cell line" OR mouse OR
mice)

Search terms were iteratively refined to prioritize communication and knowledge mobilization content while reducing retrieval of laboratory, biomarker, and treatment-focused literature.

The search strategy was designed to capture sources addressing:

- the absence of effective population-based ovarian cancer screening
- Pap test misconceptions
- symptom-based diagnostic pathways
- communication or knowledge mobilization practices related to ovarian cancer detection

3.3 Study Selection

The database search yielded 1,660 records across PubMed, CINAHL, and Web of Science. After removal of 506 duplicates, 1,154 unique records were screened by title and abstract, of which 1,082 were excluded. Seventy-two reports were sought for retrieval, and all were successfully obtained. Seventy-two full-text articles were assessed for eligibility, of which 42 were excluded. Thirty sources met the inclusion criteria and were included in the review.

Title and abstract screening, full-text review, and data charting were conducted by a single reviewer using predefined inclusion and exclusion criteria aligned with the review questions.

At the title and abstract stage, studies were excluded if they did not focus on ovarian cancer, were unrelated to screening communication, symptom awareness, help-seeking behaviour, or diagnostic pathways, or focused solely on laboratory, biomarker, or treatment-based research.

At the full-text stage, articles were excluded if they did not address ovarian cancer screening limitations, Pap test misconceptions, or symptom-based detection. Articles were also excluded if they focused exclusively on treatment outcomes or biomedical mechanisms, lacked relevance to communication or knowledge mobilization practices, or did not provide empirical or evidence-based information relevant to the review objectives.

The study selection process is presented in Figure 1.

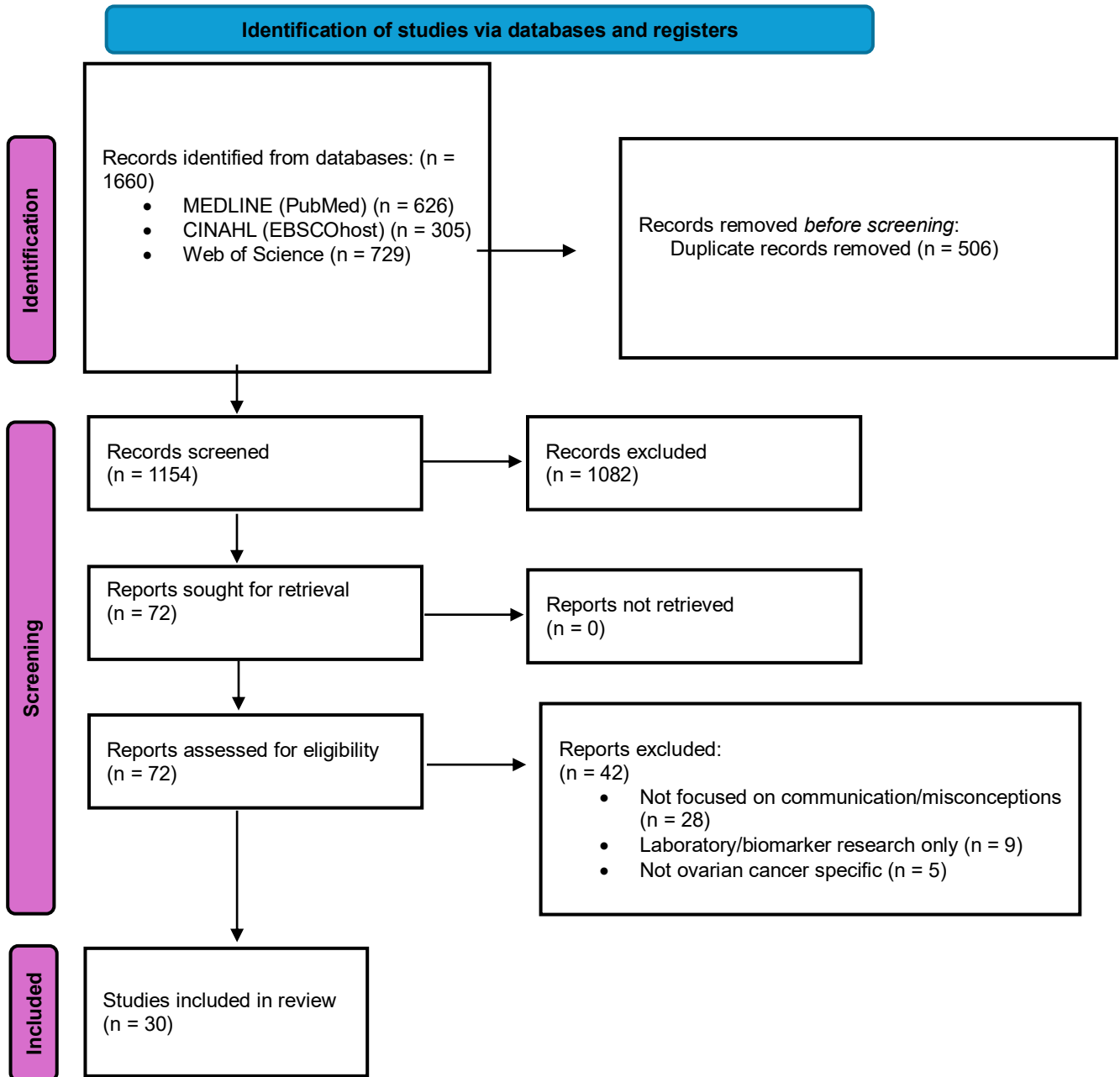


Figure 1. PRISMA-ScR flow diagram of study identification, screening, eligibility assessment, and inclusion.

3.4 Data Charting and Synthesis

Data were extracted using a standardized charting framework developed for this review. Extracted variables included author or organization, year of publication, country, source type, target audience, and key findings relevant to ovarian cancer screening communication and

knowledge mobilization. Charting was conducted by a single reviewer and refined iteratively as themes emerged to support consistency across sources.

Findings were synthesized descriptively, with attention to how the absence of routine ovarian cancer screening was framed, how Pap test misconceptions were described or addressed, how symptom-based diagnostic pathways were communicated, the types of knowledge mobilization tools identified, and distinctions between public-facing and clinician-facing communication gaps. Consistent with scoping review methodology, no formal critical appraisal of individual sources was conducted.

Data from included sources are summarized in Table 1.

Author / Organization (Year)	Country	Source Type	Target Audience	Key Findings Relevant to Knowledge Mobilization
ACOG (2017)	USA	Clinical guidance	Obstetricians and gynecologists	Confirms absence of effective screening and emphasizes symptom awareness.
Brandner et al. (2014)	Germany	Qualitative interview study	Women diagnosed with ovarian cancer (patients)	Interviews with ovarian cancer patients showed that early symptoms were often normalized or attributed to everyday causes, delaying healthcare seeking. Findings highlight the need for clearer public communication about persistent, subtle symptoms to support earlier help-seeking.
Cooper et al. (2011)	USA	Qualitative focus group study	Women aged 40–60 (general public)	Qualitative research identifies common misperceptions that the Pap test screens for ovarian cancer, contributing to false reassurance in some women.
Cooper et al. (2016)	United States	Survey study	Women in general public	Many women had low awareness of ovarian cancer symptoms, and 31.1% incorrectly believed the Pap test screens for ovarian cancer. Greater symptom awareness was associated with earlier intended care seeking, suggesting a need for clearer public education on symptoms and screening limits.
Elshami et al. (2023)	Palestine	National cross-sectional survey study	Adult women in the general public	Better ovarian cancer symptom awareness was linked to earlier intended help-seeking and fewer emotional barriers to presentation. The study suggests that educational interventions to improve symptom awareness may support earlier care-seeking.

Evans, Ziebland, and McPherson (2007)	United Kingdom	Qualitative interview study	Women diagnosed with ovarian cancer also relevant to GPs and primary care providers	Women commonly experienced delays in diagnosis due to poor symptom appraisal, normalization of vague symptoms, and provider-related delays such as non-investigation, misattribution, lack of follow-up, and referral delays. Findings support clearer public and primary care communication about ovarian cancer symptoms and the need for follow-up when symptoms persist.
Gajjar et al. (2012)	UK	Questionnaire survey of GPs	General practitioners	Symptoms are common but non-specific and difficult to interpret in primary care. Contributes to delays in investigation, showing challenges of symptom-based detection in the absence of screening.
Goff et al. (2004)	USA	Prospective case-control symptom study	Primary care clinicians, gynecologists, and researchers involved in early cancer detection and diagnostic decision-making	Research demonstrates that symptoms associated with ovarian cancer frequently overlap with common, benign conditions seen in primary care. This overlap, combined with the high prevalence of similar symptoms among women without cancer, complicates early clinical recognition and may contribute to delays in diagnostic investigation when symptom patterns are not carefully assessed.
Goldstein et al. (2015)	United States	Survey study	Women in the general public and healthcare providers	Awareness of ovarian cancer symptoms and risk factors was low among women and incomplete even among providers. Women who were more familiar with symptoms were more likely to identify them correctly and say they would report symptoms promptly, supporting the need for better public and provider education.
Gilbert et al. (2012)	Canada	Prospective pilot study	Symptomatic women aged 50 years or older	Open-access assessment of symptomatic women identified invasive ovarian cancer at a much higher rate than population screening studies, but many high-grade serous cancers were still advanced because they often originated outside the ovary. Findings suggest public and clinical messaging should emphasize prompt assessment of symptoms and the limits of current early detection approaches.

Hawkins et al. (2011)	United States	Survey study	Adult women in the general public	Although Pap test awareness and reported use were high, misconceptions about its purpose were common, with 40.6% believing it screens for ovarian cancer. Findings support clearer public education on what Pap tests do and do not detect.
Hess et al. (2012)	United States	Decision-analysis study using patient pathway data	Women with suspected ovarian cancer, also relevant to primary care and emergency clinicians	Initial provider type affected time to diagnosis, and the authors highlighted a need to improve primary care recognition of possible ovarian cancer symptoms and appropriate referral or testing. Findings support stronger clinician education around symptom interpretation and diagnostic pathways.
Holman et al. (2014)	United States	Survey study	Average-risk postmenopausal women undergoing ovarian cancer screening	Women greatly overestimated their lifetime ovarian cancer risk and showed poor understanding of the current utility of ovarian cancer screening. Findings suggest a need for better public education about actual risk and screening limitations.
Lawson-Michod et al. (2022)	United States	Qualitative interview study	Ovarian cancer patients/survivors and healthcare providers	Delays were longest during symptom appraisal and help-seeking, shaped by nonspecific symptoms, symptom normalization, embarrassment, avoidant coping, and concerns about provider judgment. Findings support better patient education on symptoms and reducing barriers to seeking care.
Liberto et al. (2022)	International (Review of global evidence)	Narrative review	Clinicians and researchers	Current and emerging biomarkers lack sensitivity and specificity for early-stage population screening.
Lockwood-Rayermann et al. (2009)	United States	Survey study	Women aged 40 and older in the general public	Awareness of ovarian cancer symptoms and risk factors was low, and more than two-thirds incorrectly believed the Pap test diagnoses ovarian cancer. Findings support the need for clearer education on ovarian cancer symptoms, risk factors, and the limits of Pap testing.
Loerzel et al. (2014)	United States	Longitudinal pre-post study	Nurse practitioner students	Knowledge deficits about ovarian cancer persisted among NP students, and many incorrectly believed the Pap test screens for ovarian cancer. Findings support stronger provider education on symptoms, risk factors, screening limits, and patient communication.

Menon et al. (2023)	UK (England, Wales, and Northern Ireland)	Randomized controlled trial	Asymptomatic, average-risk women aged 50-74 years old	No reduction in ovarian cancer mortality from CA-125 or transvaginal ultrasound screening, false-positive harms.
NICE (2015)	UK	Clinical guideline	Primary care providers	NICE recommends symptom-triggered diagnostic pathways for ovarian cancer and does not endorse population screening in average-risk, asymptomatic women.
O'Toole et al. (2024)	Ireland	Survey	General public / women	Surveys found low public confidence in recognizing ovarian cancer symptoms, with awareness remaining limited despite national awareness campaigns. Findings suggest ongoing gaps in effective public communication about ovarian cancer.
Ovarian Cancer Canada (n.d.)	Canada	Advocacy / public education	Patients and general public	Advocacy and public education materials in Canada emphasize tracking persistent and frequent symptoms and recognizing changes from a woman's baseline.
Puckett et al. (2018)	USA	Educational intervention	General public and healthcare providers	Targeted education improved understanding of Pap test limitations and ovarian cancer symptoms.
Radu et al. (2023)	UK	Cross-sectional online survey	General public (survey respondents), findings relevant to public health practitioners and policymakers	Over half of respondents incorrectly believed an ovarian cancer screening programme exists, with low symptom awareness. Indicating persistent misunderstanding of ovarian cancer detection and confusion with other gynecologic screening.
Seiback et al. (2011)	Denmark	Mixed-methods study	Women diagnosed with ovarian cancer	Women often experienced symptoms before seeking care but interpreted them as everyday bodily changes rather than signs of disease. Findings suggest a need for clearer communication about early symptoms and barriers to timely help-seeking.
Simon et al. (2012)	United Kingdom	Measurement development and validation study	General public / women, also relevant for population awareness research and intervention evaluation	The study developed validated ovarian and cervical cancer awareness measures that can identify gaps in public awareness and evaluate awareness-raising interventions. Findings support more targeted communication on ovarian cancer symptoms, risk factors, and the absence of screening.

Smits et al. (2017)	United Kingdom	Cross-sectional questionnaire study	Women at increased familial risk and women at population risk of ovarian cancer	Anticipated symptom presentation was shaped by symptom knowledge and health beliefs, with different patterns across risk groups. Findings suggest ovarian cancer information should address barriers to help-seeking and be tailored by risk level.
Stewart et al. (2012)	United States	Survey study	Primary care physicians and OB/GYNs, also women in the general public	Many physicians incorrectly believed CA-125 and transvaginal ultrasound were effective screening tests for ovarian cancer in asymptomatic, average-risk women, and public familiarity with CA-125 was also present. Findings support the need for clearer provider and public education on the lack of effective routine ovarian cancer screening and the potential harms of inappropriate testing.
Tone et al. (2022)	Canada	National patient-reported survey (quantitative with qualitative components)	Women diagnosed with ovarian cancer and received treatment in Canada	Most women reported symptoms prior to diagnosis but had limited pre-diagnostic awareness of ovarian cancer and its symptoms. Patient-reported experiences indicated different diagnostic pathways and regional differences. Frequent reports of not being taken seriously by clinicians, particularly among those with longer time to diagnosis. Findings show gaps in symptom recognition and patient-provider communication relevant to early diagnostic pathways in the absence of routine screening.
Vela-Vallespín et al. (2023)	Spain	Qualitative interview study	Women diagnosed with ovarian cancer	Delays were shaped by poor symptom recognition, false reassurance, and missed opportunities during healthcare encounters, highlighting the need for clearer symptom communication and stronger diagnostic guidance.
Jayde & Boughton (2012)	Australia	Integrative literature review	Women with possible ovarian cancer symptoms, also relevant to nurses and other healthcare providers	The review found that diagnostic delay is shaped by both patient-related and provider-related factors, including poor symptom recognition, normalization of vague symptoms, and misdiagnosis in primary care. It recommends clearer public and clinician communication about ovarian cancer symptoms and the absence of reliable screening.

Yin et al. (2024)	China	Qualitative interview study	Women diagnosed with ovarian cancer	Women often had low awareness of early ovarian cancer symptoms and cancer risk, leading to mistaken symptom attribution and delayed help-seeking. Findings support clearer public education on symptoms and stronger support for practical and psychosocial barriers to care seeking.
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Table 1. Characteristics of Included Sources Relevant to Knowledge Translation in Ovarian Cancer Detection

4 Summary of Findings

4.1 Consensus on the Absence of Routine Ovarian Cancer Screening

Across the included clinical guidance and empirical literature, there was consistent agreement that routine ovarian cancer screening should not be used in asymptomatic, average-risk individuals. Major guidance documents and large screening studies reported that transvaginal ultrasound and CA-125-based approaches do not reduce ovarian cancer mortality in this population (Menon et al., 2023). These approaches may cause harm through false-positive results, unnecessary surgery, and other downstream interventions (Liberto et al., 2022).

At the same time, several included studies showed that this absence of recommended screening is not well understood by the public. Some participants believed ovarian cancer screening was already part of routine gynecologic care (Cooper et al., 2011). Others assumed a screening program existed at the population level (Radu et al., 2023). These findings suggest that the communication gap is not only the absence of screening itself, but also persistent misunderstanding of what routine women’s health care does and does not detect.

4.2 Persistent Misunderstanding Regarding Pap Testing

Across qualitative, survey, and educational studies, misunderstanding about the purpose of the Pap test remained a recurring finding. Participants in a multisite U.S. focus group study described believing that Pap testing screened for multiple gynecologic cancers, including ovarian cancer (Cooper et al., 2011). Survey studies similarly found that many women incorrectly believed Pap tests detect ovarian cancer or that routine gynecologic care includes ovarian cancer screening (Lockwood-Rayermann et al., 2009).

Educational evidence suggests that some misconceptions can be improved, but not always fully corrected. In one intervention study, most participants correctly identified after the session that the Pap test does not screen for ovarian cancer, yet other areas of knowledge remained

inconsistent, indicating that single-session education may improve selected messages without producing broader understanding across related topics (Puckett et al., 2018). This pattern was also reflected in wider public awareness research. In a UK survey, more than half of respondents incorrectly believed that an ovarian cancer screening programme exists (Radu et al., 2023).

4.3 Emphasis on Symptom-Based Evaluation

In both public-facing resources and clinical guidance, ovarian cancer detection was framed as symptom-based rather than screening-based. Rather than recommending routine testing in asymptomatic, average-risk individuals, guidance emphasized clinical evaluation when symptoms were new, persistent, frequent, or represented a change from baseline. CA-125 testing and imaging were positioned within diagnostic pathways rather than as population screening tools (NICE, 2015).

This emphasis is especially important because several included studies showed that symptom awareness was often limited. Survey research found low recognition of ovarian cancer symptoms, including gastrointestinal symptoms (Radu et al., 2023). Educational findings suggested that symptom-related knowledge did not improve evenly across all topics (Puckett et al., 2018). Canadian patient-reported data reinforced this pattern by showing that limited symptom awareness was accompanied by delayed recognition and, in some cases, dismissive early healthcare encounters (Tone et al., 2022).

4.4 Knowledge Mobilization Tools Identified Across Sources

Knowledge mobilization tools appeared in a wide range of formats across the included sources. These included public-facing webpages, myth and fact content, symptom checklists, downloadable educational materials, structured educational sessions, awareness campaigns, clinician guidance documents, and awareness measurement tools. Advocacy and cancer organization materials primarily used plain-language content focused on symptom awareness, help-seeking prompts, and clarifications (ACOG, 2017). Clinical and professional sources more often provided clinician-facing guidance on symptom-triggered investigation and referral pathways (Simon et al., 2012). These sources also emphasized the limited role of CA-125 and imaging outside diagnostic pathways (Puckett et al., 2018). Educational and nursing-focused literature further supported the need for improved clinician knowledge and communication regarding ovarian cancer detection (Loerzel et al., 2015).

Some included studies evaluated more active educational strategies rather than static information alone. Puckett et al. (2018) found that educational sessions using CDC Inside Knowledge materials improved recognition that Pap testing does not screen for ovarian cancer and increased confidence discussing gynecologic cancer among women and providers. Loerzel et al. (2015) similarly showed that a focused awareness intervention improved ovarian cancer knowledge among nurse practitioner students, although some misconceptions persisted over time, particularly regarding Pap testing. Simon et al. (2012) contributed a different type of tool by developing and validating the Ovarian Cancer Awareness Measure, which can be used to assess symptom knowledge, awareness of risk factors, anticipated help-seeking, and public understanding that no screening programme exists.

Despite this range of tools, communication across sources was uneven. Many materials emphasized symptom awareness and advised seeking care when symptoms were new, persistent, frequent, or different from normal. However, fewer clearly explained how symptom-based diagnostic evaluation differs from routine screening. Likewise, while some resources explicitly addressed Pap test misconceptions, others focused on symptoms without directly clarifying what routine gynecologic care does and does not detect. Overall, few sources appeared to bring together, in one accessible format, a plain-language explanation of why routine ovarian cancer screening is not recommended, how ovarian cancer symptoms should be acted on, and how ovarian cancer detection differs from cervical cancer screening.

5 Discussion

5.1 Interpretation of Findings

This scoping review found a clear distinction between what is well established in clinical and research literature and what appears to be understood in public-facing contexts. Across guideline sources and empirical studies, the absence of routine ovarian cancer screening for asymptomatic, average-risk individuals was consistently presented as settled evidence, with symptom-triggered evaluation positioned as the appropriate pathway to investigation. Misunderstanding among the public remained common across survey, qualitative, and patient-reported studies. This included confusion about the purpose of the Pap test, belief that an ovarian cancer screening programme exists, and uncertainty about how symptoms should be interpreted or acted on.

Taken together, these findings suggest that the central issue is not lack of evidence, but lack of communicative alignment across audiences. Messages that are clear in guideline language do not necessarily become clear in public understanding. Public-facing communication often emphasizes symptom vigilance, but does not always explain why screening is unavailable, how ovarian cancer detection differs from cervical cancer screening, or what symptom-based evaluation may actually involve. As a result, the evidence base appears more coherent than the communication environment surrounding it.

5.2 Identified Knowledge Mobilization Gaps

This review identified several knowledge mobilization gaps that extended beyond simple lack of awareness. First, explanations distinguishing Pap testing, ovarian cancer screening, and symptom-triggered diagnostic evaluation were not consistently integrated across sources. Although some materials directly corrected Pap test misconceptions, misunderstanding remained common among the public (Cooper et al., 2011). Similar gaps in understanding were also reported among healthcare learners (Hawkins et al., 2011). Misconceptions were further observed among providers, indicating that these knowledge gaps are not limited to patient populations (Stewart et al., 2012). Second, the absence of routine ovarian cancer screening was often stated, but less often translated into clear, plain-language guidance about what women should do in practice when symptoms arise. This is important because studies have shown that

low symptom awareness and normalization of bodily changes contribute to delays in care (Brandner et al., 2014; Vela-Vallespín et al., 2023). False reassurance and uncertainty about when to seek care have also been identified as contributing factors in delayed help-seeking (Lawson-Michod et al., 2022).

A further gap concerned the uneven alignment between public-facing and clinician-facing communication. Public resources commonly emphasized symptom vigilance, whereas clinician-oriented sources focused more on investigation and referral pathways (Gajjar et al., 2012). However, few sources appeared to connect these two levels in a way that clearly supported patient action and provider response within the same communication pathway. Studies involving primary care clinicians and other providers suggest that gaps in ovarian cancer knowledge are not limited to the public, particularly in relation to symptom interpretation (Goldstein et al., 2017). Gaps in timely follow-up and referral practices have also been reported (Hess et al., 2012). Together, these findings suggest that symptom awareness alone is insufficient unless it is paired with explicit explanation of screening limitations, clearer guidance on symptom appraisal and help-seeking, and stronger coherence between public-facing and clinician-facing messages.

5.3 Differentiation Between Public-Facing and Clinician-Facing Communication Gaps

An important finding of this review was that public-facing and clinician-facing communication gaps were related, but not identical. Public-facing gaps were mainly problems of clarity, explicitness, and usability. Advocacy and educational materials often encouraged attention to symptoms, but did not always clearly explain what the absence of routine ovarian cancer screening means in practice, how ovarian cancer detection differs from cervical cancer screening, or when persistent symptoms should prompt further medical follow-up. This is consistent with studies showing ongoing public confusion about Pap testing and ovarian cancer screening (Cooper et al., 2011). Misunderstanding has also been reported in relation to the purpose and scope of cervical screening (Hawkins et al., 2011). Additional work highlights limited awareness of symptom significance and interpretation (Lockwood-Rayermann et al., 2009). More recent evidence further demonstrates persistent uncertainty surrounding vague or gastrointestinal symptoms (Radu et al., 2023).

Clinician-facing gaps, in contrast, appeared less related to absence of evidence than to uneven application and communication of that evidence in real-world care. Although guidelines describe symptom-triggered investigation pathways, studies suggest that provider knowledge and response are not always consistent, particularly in relation to symptom interpretation (Gajjar et al., 2012). Inconsistent reassurance following negative or non-specific findings has also been reported (Goldstein et al., 2017). Delays in timely follow-up or referral further indicate variability in clinical response (Hess et al., 2012). Gaps in knowledge and application have also been observed among healthcare learners and providers more broadly (Stewart et al., 2012).

Qualitative and patient-reported studies further show that women sometimes feel dismissed or that their symptoms are misattributed during early healthcare encounters (Lawson-Michod et al., 2022). Patients have also reported insufficient guidance when navigating symptoms and care pathways (Tone et al., 2022). Similar findings highlight challenges in patient-provider communication during early diagnostic interactions (Vela-Vallespín et al., 2023). Earlier work

likewise documented experiences of dismissal and delayed recognition in clinical encounters (Evans et al., 2007).

Viewed together, these findings suggest that the challenge is not only to improve public awareness or clinician knowledge separately, but to improve coherence between them. Public-facing communication must better prepare women to interpret and act on symptoms, while clinician-facing resources must better support consistent explanation, investigation, and follow-up when those concerns are brought forward.

5.4 Implications for Knowledge Mobilization

The findings of this review suggest that the main knowledge mobilization problem is the fragmentation of message delivery across audiences, formats, and points of care. Included sources identified two broad communication streams: public-facing materials focused on symptom awareness, myths, and help-seeking, and clinician-facing guidance focused on investigation, referral, and diagnostic management. What was less visible across the literature was communication that clearly linked these two levels. In particular, few tools appeared to connect what women are told to notice and act on with what primary care clinicians are expected to explain, investigate, or follow up when those concerns are raised.

These findings suggest that effective knowledge mobilization in this area requires more than increasing awareness of symptoms alone. Public-facing communication may be stronger when it clearly distinguishes the roles and limitations of different forms of gynecologic care, explains what is and is not routinely detected, and provides actionable guidance on how to respond to potential signs of concern. This is especially relevant given the recurring findings of Pap test confusion, low symptom awareness, false reassurance, and uncertainty about when to seek care across survey, qualitative, and patient-reported studies.

The review also suggests that clinician-facing knowledge mobilization could be improved through tools that support communication, not only clinical decision-making. Although guideline documents describe symptom-triggered pathways, studies indicate that provider knowledge and interpretation are not always consistent in practice (Gajjar et al., 2012). Variability in follow-up and referral patterns has also been reported (Hess et al., 2012). Inconsistent application of evidence and gaps in clinical understanding have been observed among healthcare providers more broadly (Stewart et al., 2012). Challenges in clinical decision-making and diagnostic processes have also been documented (Goldstein et al., 2017). Structured conversation aids, consultation prompts, brief scripts, or integrated FAQ-style tools may therefore be useful in helping clinicians explain why screening is not recommended, what symptoms warrant reassessment, and how uncertainty should be managed over time rather than resolved through reassurance alone.

Timing is also important. Many educational resources are encountered after diagnosis, specialist referral, or entry into cancer care systems, whereas the key intervals identified in several included studies occur much earlier, during symptom appraisal, normalization, initial help-seeking, and first-line healthcare encounters. This suggests that knowledge mobilization efforts

may be most useful when positioned earlier in the care continuum, particularly in community, primary care, and general women's health contexts.

Collectively, this review brings together clinical guidance, awareness research, patient experience evidence, and existing communication tools within a single knowledge mobilization framework. By doing so, it helps identify not only what information is available, but where communication remains incomplete, poorly aligned, or insufficiently actionable across public-facing and clinician-facing settings.

5.5 Strengths and Limitations

This review has several strengths. A scoping review approach was well suited to the aim of mapping how ovarian cancer detection is communicated in the absence of routine population screening across a heterogeneous body of evidence. The inclusion of multiple source types, including clinical guidance, trials, qualitative studies, surveys, educational interventions, and grey literature, allowed the review to examine not only what evidence exists, but how that evidence is translated across public-facing and clinician-facing contexts. A further strength is the review's focus on knowledge mobilization rather than clinical effectiveness alone, which helped identify differences between evidence availability, public understanding, and communication practices.

This review also has limitations. Although the search strategy was structured, it was targeted rather than fully exhaustive and may not have captured all relevant sources, particularly within grey literature and non-indexed communication materials. The inclusion of diverse evidence types supported breadth but limited direct comparison across studies and resources. Grey literature searching also presents challenges for reproducibility because website content, campaign materials, and advocacy resources may change over time or be difficult to systematically retrieve. In addition, the review focused primarily on English-language sources and was weighted toward high-income settings, which may limit transferability to other linguistic, cultural, or health system contexts. Non-English and non-high-income country sources were excluded pragmatically given the scope and resources of this review; however, this decision limits the generalizability of findings and may have excluded relevant communication approaches from other health system contexts. In addition, screening, full-text review, and data charting were conducted by a single reviewer, which may have increased the risk of selection bias or interpretive subjectivity, reduced confidence in study inclusion decisions, and limits credibility for some journals. While single-reviewer screening is not uncommon in scoping reviews, this limitation was not formally mitigated through calibration exercises or audit checks. A further limitation concerns the depth of knowledge mobilization analysis. Although KM tools were catalogued across included sources, the review did not systematically extract whether KM strategies were formally evaluated. Evaluation of dissemination outcomes was rare across included sources, and few studies compared passive versus active knowledge mobilization approaches or reported outcomes beyond knowledge change, such as behaviour change, help-seeking, or system response. A more explicit extraction of evaluation evidence, even to confirm its absence, would have strengthened the KM contribution of this review. Finally, consistent with scoping review methodology, the review was designed to map concepts, communication

patterns, and resource gaps rather than formally appraise study quality or determine the effectiveness of individual knowledge mobilization tools.

References

- American College of Obstetricians and Gynecologists. (2017). The role of the obstetrician-gynecologist in the early detection of epithelial ovarian cancer. *ACOG Committee Opinion*, 716. Retrieved from:

<https://www.acog.org/-/media/project/acog/acogorg/clinical/files/committee-opinion/articles/2017/09/the-role-of-the-obstetrician-gynecologist-in-the-early-detection.pdf?rev=12a7f0f4a69d44708fcbd9e36701bb7c&hash=72D6B1F1629148C592A7DA89F06BF820>
- Arksey, H., & O'Malley, L. (2005). Scoping studies: Towards a methodological framework. *International Journal of Social Research Methodology*, 8(1), 19–32.

doi: <https://doi.org/10.1080/1364557032000119616>
- Cooper, C. P., Polonec, L., & Gelb, C. A. (2011). *Women's Knowledge and Awareness of Gynecologic Cancer: A Multisite Qualitative Study in the United States*. *Journal of Women's Health*, 20(4), 517–524.

doi: <https://doi.org/10.1089/jwh.2011.2765>
- Cooper, C. P., Gelb, C. A., Trivers, K. F., & Stewart, S. L. (2016). Intended care seeking for ovarian cancer symptoms among U.S. women. *Preventive medicine reports*, 3, 234–237.

doi: <https://doi.org/10.1016/j.pmedr.2016.01.011>

Elshami, M., Al-Madhoun, S., Alser, M., Al-Slaibi, I., Yaseen, A., Tuffaha, A., Jabr, H., Ubaiat,

S., Khader, S., Khraishi, R., Jaber, I., Arafeh, Z. A., Alqattaa, A., Hadi, A. A. E.,

Barhoush, O., Hijazy, M., Eleyan, T., Alser, A., Hziema, A. A., Shatat, A., Bottcher,

B. (2023). Anticipated time to seek medical advice for possible ovarian cancer symptoms

and perceived barriers to early presentation among Palestinian women: a national cross-

sectional study. *BMC cancer*, 23(1), 975.

doi: <https://doi.org/10.1186/s12885-023-11484-z>

Evans, J., Ziebland, S., & McPherson, A. (2007). Minimizing delays in ovarian cancer diagnosis:

an expansion of Andersen's model of 'total patient delay'. *Family practice*, 24(1), 48–55.

doi: <https://doi.org/10.1093/fampra/cml063>

Gajjar, K., Ogden, G., Mujahid, M. I., & Razvi, K. (2012). *Symptoms and Risk Factors of*

Ovarian Cancer: A Survey in Primary Care. Obstetrics and Gynecology, 2012.

doi: <https://doi.org/10.5402/2012/754197>

Gilbert, L., Basso, O., Sampalis, J., Karp, I., Martins, C., Feng, J., Piedimonte, S., Quintal, L.,

Ramanakumar, A. V., Takefman, J., Grigorie, M. S., Artho, G., Krishnamurthy, S., &

DOvE Study Group (2012). Assessment of symptomatic women for early diagnosis of

ovarian cancer: results from the prospective DOvE pilot project. *The Lancet. Oncology*, 13(3), 285–291.

doi: [https://doi.org/10.1016/S1470-2045\(11\)70333-3](https://doi.org/10.1016/S1470-2045(11)70333-3)

Goff, B.A., Mandel L.S., Melancon, C.H., & Muntz, H.G. (2004). Frequency of Symptoms of Ovarian Cancer in Women Presenting to Primary Care Clinics. *JAMA*, 291

doi: 10.1001/jama.291.22.2705

Goldstein, C.L., Sheeder, J., Medlin, E.E., Cullen, P.L., Hyman, D., & Behbakht, K. (2017).

Ovarian Cancer Knowledge Among Advanced Providers in a University Setting. *The Journal for Nurse Practitioners*, 13.

doi: [10.1016/J.NURPRA.2016.12.014](https://doi.org/10.1016/J.NURPRA.2016.12.014)

Hawkins, N. A., Cooper, C. P., Saraiya, M., Gelb, C. A., & Polonec, L. (2011). Why the Pap

test? Awareness and use of the Pap test among women in the United States. *Journal of women's health* (2002), 20(4), 511–515.

doi: <https://doi.org/10.1089/jwh.2011.2730>

Hess, L. M., Stehman, F. B., Method, M. W., Weathers, T. D., Gupta, P., & Schilder, J. M.

(2012). Identification of the optimal pathway to reach an accurate diagnosis in the absence of an early detection strategy for ovarian cancer. *Gynecologic oncology*, 127(3), 564–568.

doi: <https://doi.org/10.1016/j.ygyno.2012.08.029>

Holman, L. L., Lu, K. H., Bast, R. C., Jr, Hernandez, M. A., Bodurka, D. C., Skates, S., & Sun,

C. C. (2014). Risk perception, worry, and test acceptance in average-risk women who undergo ovarian cancer screening. *American Journal of Obstetrics and Gynecology*, 210(3), 257.e1–257.e2576.

doi: <https://doi.org/10.1016/j.ajog.2013.11.022>

Jayde, V & Boughton, M. (2012) The diagnostic journey of ovarian cancer: A review of the literature and suggestions for practice. *Contemporary Nurse*, 41(1) 5-17.

doi: 10.5172/conu.2012.41.1.5

Lawson-Michod, K. A., Watt, M. H., Grieshober, L., Green, S. E., Karabegovic, L., Derzon, S.,

Owens, M., McCarty, R. D., Doherty, J. A., & Barnard, M. E. (2022). Pathways to ovarian cancer diagnosis: a qualitative study. *BMC women's health*, 22(1), 430.

doi: <https://doi.org/10.1186/s12905-022-02016-1>

Liberto, J. M., Chen, S.-Y., Shih, I.-M., Wang, T.-H., Wang, T.-L., & Pisanic, T. R., II. (2022).

Current and Emerging Methods for Ovarian Cancer Screening and Diagnostics: A Comprehensive Review. Cancers, 14(12), 2885.

doi: <https://doi.org/10.3390/cancers14122885>

Lockwood-Rayermann, S., Donovan, H. S., Rambo, D., & Kuo, C. W. (2009). Women's awareness of ovarian cancer risks and symptoms. *The American Journal of Nursing*, 109(9), 36–46.

doi: <https://doi.org/10.1097/01.NAJ.0000360309.08701.73>

Loerzel, V. W., Hunt, D., & Rash, E. (2015). A Pap test does not screen for everything: nurse practitioner knowledge of ovarian cancer. *Journal of the American Association of Nurse Practitioners*, 27(3), 124–130.

doi: <https://doi.org/10.1002/2327-6924.12151>

Menon, U., Gentry-Maharaj, A., Burnell, M., Ryan, A., Kalsi, J. K., Singh, N., Dawney, A., Fallowfield, L., McGuire, A. J., Campbell, S., Skates, S. J., & Parmar, M., & Jacobs, I. J. (2023). Mortality impact, risks, and benefits of general population screening for ovarian cancer: the UKCTOCS randomised controlled trial. *Health Technol Assess*.

doi: <https://doi.org/10.3310/BHBR5832>

National Institute for Health and Care Excellence. (2015). *Suspected cancer: Recognition and referral (NG12)*. Retrieved from:

<https://www.nice.org.uk/guidance/ng12>

Ovarian Cancer Canada. (n.d.). *Symptoms and detection*. Retrieved from:

<https://ovariancanada.org/symptoms-detection>

Puckett, M. C., Townsend, J. S., Gelb, C. A., Hager, P., Conlon, A., & Stewart, S.L. (2018).

Ovarian Cancer Knowledge in Women and Providers Following Education with Inside Knowledge Campaign Materials. *J Cancer Educ*, 33(6), 1285-1293.

doi: 10.1007/s13187-017-1245-0

Radu, C.-A., Fernandes, N. M. de M., Khalfe, S., & Stordal, B. (2023). *Awareness of Ovarian Cancer Symptoms and Risk Factors in a Young Ethnically Diverse British Population. Cancer Medicine*, 12(8), 9879-9892.

doi: <https://doi.org/10.1002/cam4.5670>

S. Brandner., Müller-Nordhorn, J., Stritter, W., Christina, F., Sehouli, J., & Holmberg, C.

(2014). Symptomization and triggering processes: Ovarian cancer patients' narratives on pre-diagnostic sensation experiences and the initiation of healthcare seeking. *Social Science & Medicine*, 119.

doi: <https://doi.org/10.1016/j.socscimed.2014.08.022>.

Seibaek, L., Petersen, L. K., Blaakaer, J., & Hounsgaard, L. (2011). Symptom interpretation and health care seeking in ovarian cancer. *BMC women's health*, 11, 31.

doi: <https://doi.org/10.1186/1472-6874-11-31>

Simon, A. E., Wardle, J., Grimmett, C., Power, E., Corker, E., Menon, U., Matheson, L., &

Waller, J. (2012). Ovarian and cervical cancer awareness: development of two validated

measurement tools. *The journal of family planning and reproductive health care*, 38(3), 167–174.

doi: <https://doi.org/10.1136/jfprhc-2011-100118>

Smits, S., Boivin, J., Menon, U., & Brain, K. (2017). Influences on anticipated time to ovarian cancer symptom presentation in women at increased risk compared to population risk of ovarian cancer. *BMC cancer*, 17(1), 814.

doi: <https://doi.org/10.1186/s12885-017-3835-y>

Stewart, S. L., Rim, S. H., & Gelb, C. A. (2012). Physician knowledge and awareness of CA-125 as a screen for ovarian cancer in the asymptomatic, average-risk population. *Health education & behavior : the official publication of the Society for Public Health Education*, 39(1), 57–66.

doi: <https://doi.org/10.1177/1090198111407185>

Tone, A., Boghosian, T., Ross, A., Baugh, E., Altman, A.D., Dawson, L., Reid, F., & Crawford C. (2022). Understanding the Experience of Canadian Women Living with Ovarian Cancer through the Every Woman Study™. *Curr. Oncol.*, 29, 3318–3340.

doi: <https://doi.org/10.3390/currenol29050271>

Tricco, A. C., Lillie, E., Zarin, W., O'Brien, K. K., Colquhoun, H., Levac, D., Moher, D., Peters,

M. D. J., Horsley, T., Weeks, L., Hempel, S., Akl, E. A., Chang, C., McGowan, J.,

Stewart, L., Hartling, L., Aldcroft, A., Wilson, M. G., Garritty, C., Straus, S. E. (2018).

PRISMA extension for scoping reviews (PRISMA-ScR): Checklist and

explanation. *Annals of Internal Medicine*, 169(7), 467–473.

doi: <https://doi.org/10.7326/M18-0850>

University Health Network. (n.d.). *Ovarian cancer*. UHN Gynecologic Oncology Program.

Retrieved from: <https://gynoncuhn.ca/ovarian-cancer/>

Vela-Vallespín, C., Medina-Perucha, L., Jacques-Aviñó, C., Codern-Bové, N., Harris, M.,

Borras, J. M., & Marzo-Castillejo, M. (2023). Women's experiences along the ovarian

cancer diagnostic pathway in Catalonia: A qualitative study. *Health expectations : an*

international journal of public participation in health care and health policy, 26(1), 476–

487.

doi: <https://doi.org/10.1111/hex.13681>

Yin H, Zha Y, Zhou Y, Tao, H., & Zhu, D. (2024). What are the barriers and facilitators to help-

seeking behaviour for symptoms in patients with ovarian cancer in China? A qualitative

study. *BMJ open*, 14(11).

doi:10.1136/mjopen-2024-087602